

Map to ROI: Hierarchical Media Mix Modeling

Understanding Regional Differences in Advertising Effectiveness

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1 Introduction

The progression of marketing decision-making has been characterized by an increasing dependence on sophisticated analytics, enabling organizations to gain deeper insights into the effectiveness of advertising expenditures. Media Mix Modeling (MMM) plays a critical role in this process by quantifying the contribution of individual media channels to business outcomes and informing optimal budget allocation decisions. While conventional MMM approaches are widely adopted, they often rely on national-level aggregation or simplified linear assumptions, which can obscure meaningful regional heterogeneity and fail to capture realistic advertising behaviors such as carryover effects and diminishing returns.

Local markets differ substantially in demographic composition, economic conditions, lifestyle characteristics, population density, and media consumption patterns. These differences influence both baseline sales levels and consumer responsiveness to advertising. Treating all regions as homogeneous can therefore lead to biased parameter estimates, unreliable recommendations, and inefficient allocation of marketing resources. Furthermore, advertising effects do not manifest instantaneously nor increase indefinitely; instead, they accumulate over time and eventually saturate. Accurately modeling these dynamics requires methodologies that reflect real-world consumer response behavior.

This project addresses these challenges through the development of a hierarchical Bayesian Media Mix Model (MMM) that integrates geo-level structure, nonlinear advertising response functions, and contextual factors within a unified probabilistic framework. The hierarchical Bayesian formulation enables partial pooling across regions, allowing markets with similar characteristics to share information while preserving meaningful local differences. This approach improves parameter stability, reduces estimation variance, and yields more reliable media effectiveness estimates, particularly in markets with sparse or irregular data.

The methodological framework incorporates realistic media transformations, including adstock functions to model carryover effects and Hill-type saturation functions to capture diminishing returns at higher spending levels. These components are essential for producing credible ROI estimates and for simulating the impact of budget reallocation on future performance. Additionally, temporal structures such as trends and seasonality are explicitly modeled to separate media-driven effects from broader market dynamics, enhancing interpretability and alignment with business patterns.

By combining contextual features, advanced Bayesian inference, and a structured representation of geographic heterogeneity, this project provides Butler/Till with a robust analytical framework for evaluating media performance. The resulting model enables more granular, region-specific budgeting strategies, improved scenario analysis, and enhanced forecasting accuracy. In an environment where accountability and precision are increasingly critical, this geo-aware probabilistic approach represents a substantial advancement in media measurement and marketing effectiveness analysis.

1.1 Problem Statement

In today’s fragmented media ecosystem, advertisers allocate substantial budgets across multiple channels—including linear television, digital video, social media, and display advertising—while operating across diverse geographic markets. Understanding the true return on investment (ROI) for each channel within each market is essential for effective media optimization. However, traditional marketing attribution approaches face several critical challenges:

- **Geographic heterogeneity:** Media effectiveness varies significantly across Designated Market Areas (DMAs) due to differences in demographics, consumer behavior, and market maturity.
- **Sparse and unbalanced data:** Many DMAs exhibit inconsistent media activity or limited observations, complicating reliable estimation.
- **Complex response patterns:** Media effects exhibit carryover (adstock), saturation (diminishing returns), and seasonal variation that must be explicitly modeled.
- **Attribution complexity:** Sales outcomes are influenced by multiple concurrent factors, making it difficult to isolate the causal impact of any single channel.

1.2 Project Goals

This project develops a hierarchical Bayesian Marketing Mix Model (MMM) to quantify the causal relationship between media spending and revenue across 169 DMAs. The primary objectives are as follows:

- Estimate media effectiveness at both the cluster and DMA levels for four primary channels: linear television, digital video, social media, and display advertising.
- Account for geographic heterogeneity through clustering and hierarchical modeling while enabling information sharing across similar markets.
- Handle missing data systematically using imputation strategies and probabilistic inference.
- Provide actionable insights for media planning by quantifying ROI and optimal spending levels by channel and market.

1.3 Approach and Impact

We employ a two-stage modeling approach. First, DMAs are grouped into homogeneous clusters based on contextual attributes such as population density, income, and age distribution. Second, a hierarchical Bayesian model is constructed to estimate media effects at the cluster level while allowing DMA-specific deviations. This approach enables:

- More stable parameter estimates for smaller or data-sparse DMAs through partial pooling.
- Interpretable patterns that reveal how different market archetypes respond to media investments.
- Full probabilistic quantification of uncertainty across all estimates.
- Scalable inference using modern Bayesian computation frameworks (PyMC with JAX).

2 Dataset Description

This study integrates multiple data sources to support geo-level media mix modeling and contextual analysis. Specifically, the dataset comprises (i) ZIP-code-level demographic and socioeconomic attributes and (ii) weekly marketing spend and sales data aggregated at the Designated Market Area (DMA) level. Together, these sources provide the basis for all preprocessing, feature engineering, and subsequent modeling steps.

2.1 Contextual Demographic Data (Raw CX Dataset)

The contextual dataset contains detailed demographic, economic, and housing characteristics at the ZIP-code level. These variables are used to characterize regional heterogeneity and to inform clustering and hierarchical pooling strategies within the modeling framework. Table 1 summarizes the primary categories of variables included.

Table 1: ZIP-Code Level Contextual Demographic and Socioeconomic Variables

Category	Variables Description
Geographic Identifiers	ZIP code and name, state code and name, county code and name, DMA code and name
Population and Household Composition	Total population, population aged 18 and over, population under 18, family population, total households, family households, non-family households, average household size, average family size, households with children, single-person households
Economic and Income Characteristics	Aggregate household income, average household income, median household income, per-capita income, estimated disposable income, households below the poverty line, poverty rate
Age and Labor Market Indicators	Median age, working-age population, total labor force, labor force participation rate, employed population, unemployed population, unemployment rate
Housing, Education, and Urbanization	Total housing units, owner-occupied housing, renter-occupied housing, homeownership rate, educational attainment (high school or higher; bachelor’s degree or higher), urban–rural indicator

2.2 Marketing Spend and Sales Data (Raw DMA–Week Panel)

The expenditure on marketing and sales figures constitute a weekly panel at the Designated Market Area (DMA) level, serving as the principal temporal input for the modeling pipeline. Each row represents a distinct combination of DMA and week, characterized by identifiers including *DMA name*, *DMA code*, *week start date*, *week end date*, and *year–week*. These identifiers collectively delineate the geographic scope and temporal alignment of the observations.

Investments in paid media are documented as raw weekly expenditure figures across multiple channels, including *PM spend TV linear*, *PM spend video*, *PM spend social*, *PM spend display*, *PM spend search*, *PM spend audio*, and *PM OOH events*. The variables presented here represent direct media

expenditures as reported by upstream systems and have not been adjusted for inflation, delayed carryover effects, saturation behavior, or cross-channel interactions at this stage of the pipeline.

The dependent variable, *DV sales*, reflects observed weekly sales at the DMA level in its untransformed form, without normalization, variance stabilization, or seasonal adjustment. Collectively, these variables form the unaltered foundation for all subsequent preprocessing and feature engineering procedures, including adstock modeling, nonlinear response curve estimation, temporal smoothing, and the integration of aggregated contextual demographic features derived from ZIP code-level data.

3 Missing Data and Imputation

3.1 Missingness Patterns

A thorough assessment of missingness was conducted following the creation of a complete DMA-week panel, ensuring that unobserved periods were explicitly included rather than implicitly excluded during the preprocessing stage.

The dependent variable, *dv sales*, exhibits missing values primarily attributable to reporting delays and incomplete weekly submissions, particularly toward the end of the observation period. Variables associated with paid media expenditure, including *pm spend tv linear*, *pm spend video*, *pm spend social*, *pm spend display*, and other relevant channels, display missingness patterns that predominantly coincide with weeks characterized by zero or inactive campaign activity. This behavior suggests a structural origin of the missing data rather than a stochastic one.

Contextual demographic variables, collectively denoted as *cx*, exhibit substantially lower rates of missingness once aggregated from ZIP-code to DMA-level geography. Residual gaps in these features are primarily associated with sparsely populated ZIP codes, suppressed census estimates, or zero-valued denominators arising during aggregation, rather than reflecting systematic data quality concerns.

Intermittent missingness in the out-of-home activity indicator, *pm ooh events*, corresponds to periods during which no qualifying events were documented, consistent with the operational definition of the variable.

3.2 Imputation Methods

For paid media expenditure variables, all missing values were imputed as zero following construction of the complete DMA-week panel. This choice is supported by the non-negative nature of media spending and empirical evidence indicating that missing values predominantly correspond to periods of inactive or absent campaigns rather than reporting errors.

Contextual demographic variables that are time-invariant or evolve slowly and exhibit minimal missingness were imputed using measures of central tendency. Mean imputation was applied to approximately symmetric distributions, while median imputation was used for skewed variables, particularly those related to income. Prior to aggregation, ZIP-level observations with zero population or household counts were treated as missing to prevent the introduction of artificial zeros into population-weighted averages.

Missing values in *pm ooh events* were imputed as zero and subsequently transformed into a binary indicator reflecting the presence or absence of an out-of-home event within a given DMA-week. In

contrast, missing values in the sales outcome, *dv sales*, were intentionally not imputed. Observations with absent sales values were retained within the panel structure but excluded from the likelihood contribution during model estimation.

3.3 Justification and Impact

The evaluation of the impact of the missing data treatment was conducted through a comprehensive series of diagnostic and validation checks. The distributional comparisons performed prior to and following imputation reveal no significant alterations in sales levels, media exposure patterns, or distributions of contextual features. The diagnostics at the DMA level provide confirmation that the panel maintains a wide geographic representation, while also clearly identifying DMAs that exhibit lower temporal coverage instead of omitting them without acknowledgment. Temporal sanity checks indicate that periods of zero sales, which are due to reporting lags, can be identified and excluded from the training dataset. This exclusion helps to prevent any downward bias in the most recent observations.

From a modeling perspective, the chosen methodology guarantees that all features are clearly defined, economically interpretable, and uniformly scaled, while preventing any leakage from imputation into the target variables. By clearly differentiating among structural zeros, reporting delays, and authentically absent information, the missing data framework maintains statistical integrity and facilitates the reliable estimation of media response and sales dynamics. This systematic approach to addressing missing data facilitates dependable conclusions while ensuring clarity and reproducibility across the modeling process.

4 Exploratory Analysis

Exploratory analysis was conducted to evaluate data quality, structural consistency, and relationships between contextual characteristics, media inputs, and sales outcomes prior to model development. Both the ZIP-level contextual data and DMA-week media dataset were reviewed for completeness and anomalies. Minor missingness and invalid household counts in the contextual data were corrected, and DMA-week observations with all-zero numeric values were removed. After cleaning, the datasets exhibited consistent DMA coverage.

ZIP-level contextual variables were aggregated to the DMA level using sums for population, households, and expenditures, and population-weighted averages for income, household size, and median age. Additional derived metrics—including population density, income per household, expenditure per household, and grouped alcohol expenditure—were constructed to improve interpretability. Sales were normalized by population to compute *sales per capita*, which serves as the primary efficiency metric.

Bivariate analysis indicated that sales efficiency is not driven solely by market size. While larger DMAs exhibit higher total sales, sales per capita shows stronger associations with median income, population density, and overall consumer expenditure. These relationships are illustrated in Figure 1.

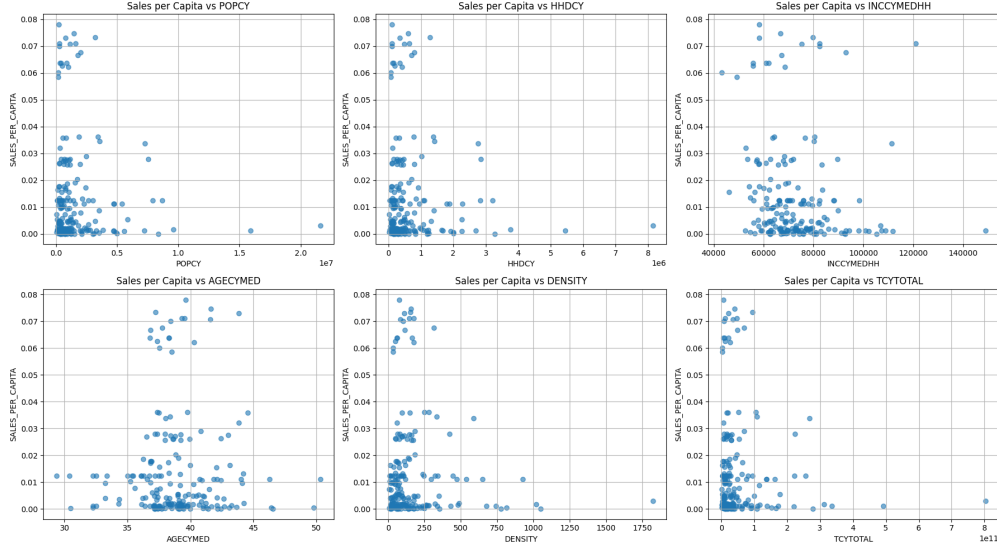
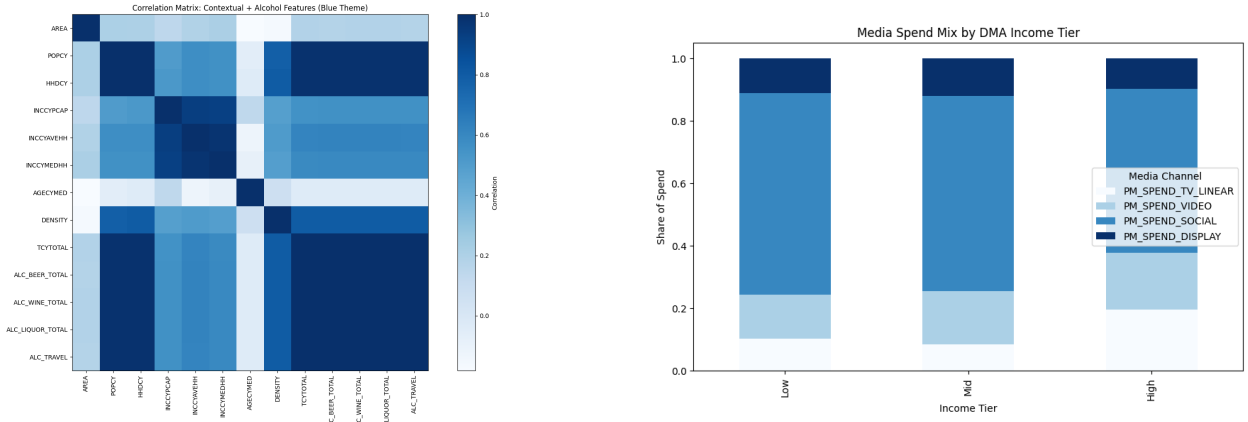


Figure 1: Sales per Capita vs Contextual Variables

Correlation analysis further highlighted strong collinearity among income- and expenditure-related variables, while population density emerged as a largely independent explanatory dimension. These relationships are summarized in Figure 2, which presents a correlation heatmap of contextual and alcohol-related features.



(a) Correlation Heatmap

(b) Media Spend Mix by Income Tier

Figure 2: Contextual and media allocation patterns across DMAs. (a) Correlation heatmap illustrating clustering among income, expenditure, population, and alcohol-related contextual features. (b) Distribution of media spend shares across DMA income tiers, highlighting systematic differences in channel mix by market affluence.

Finally, media allocation patterns differed systematically across income tiers. Higher-income DMAs allocated a larger proportion of spend to Video and Social channels, while lower-income DMAs relied more heavily on Linear Television. These differences in channel mix are summarized in Figure ??.

Beyond static contextual relationships, we examined national-level temporal dynamics in paid media

activity and their association with sales outcomes. Understanding these dynamics is critical for motivating the inclusion of lagged and nonlinear media transformations in subsequent modeling stages.

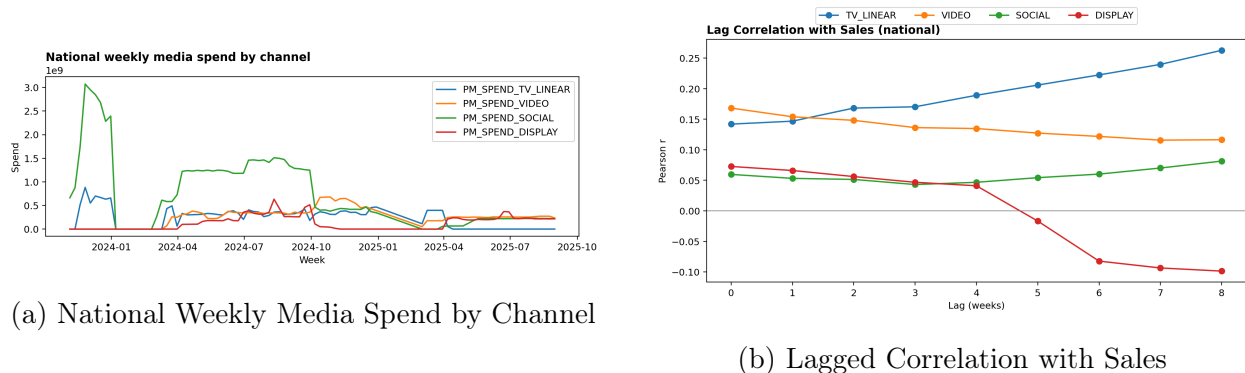


Figure 3: National media dynamics and sales association. (a) Weekly paid media expenditure by channel, illustrating scale differences and volatility patterns. (b) Lagged Pearson correlations between media spend and sales, motivating the inclusion of adstock and delayed response effects.

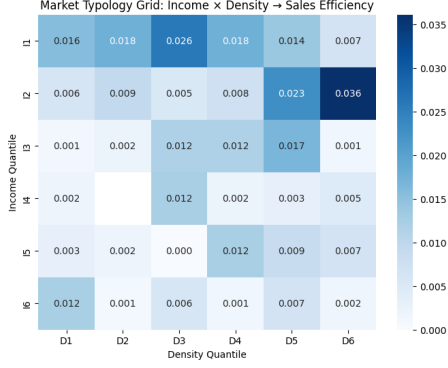
Figure 3 illustrates two complementary perspectives. Panel (a) shows weekly national media spend by channel, highlighting substantial heterogeneity in scale and volatility across channels. Social and Video exhibit pronounced bursts and regime shifts, while Linear TV displays comparatively steadier spending patterns. These dynamics suggest that instantaneous attribution is insufficient and that persistence effects must be explicitly modeled.

Panel (b) presents lagged Pearson correlations between national sales and media spend across multiple weekly offsets. Linear TV demonstrates increasing correlation strength at longer lags, consistent with delayed response and carryover effects. Video shows moderate short-term correlation that attenuates with lag, while Social exhibits weaker but gradually increasing association. Display correlations decay rapidly and turn negative at longer lags, suggesting limited persistence. Together, these patterns provide empirical justification for incorporating adstock and channel-specific response dynamics in the hierarchical Bayesian model.

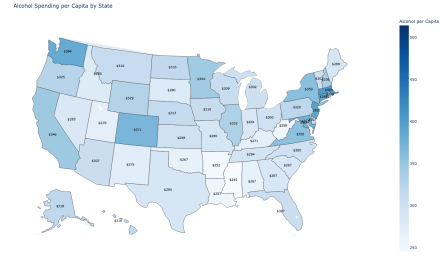
To further segment markets, DMAs were categorized using a 6×6 grid defined by income and population density quantiles. Median sales per capita increased monotonically from low-income, low-density markets to high-income, high-density markets, highlighting systematic structural differences in demand across regions. This segmentation is presented in

Additional diagnostics compared each DMA’s share of total population with its share of total sales. High-income and densely populated DMAs generally outperformed relative to their population share, whereas rural, lower-income markets tended to underperform.

Given the availability of granular alcohol expenditure data, alcohol spending was evaluated as a behavioral proxy for consumption intensity. Alcohol spending per capita exhibited strong geographic clustering, with elevated levels observed in the Northeast, Pacific Northwest, and select Mountain states, while much of the South and Midwest displayed comparatively lower values. Figure 4 presents state-level alcohol expenditure per capita, normalized by population. This normalization reveals contrasts that are obscured in total-spend comparisons and aligns closely with observed variation in sales efficiency across markets.



(a) Income × Density Market Grid



(b) Alcohol Expenditure per Capita by State

Figure 4: Structural and behavioral variation across U.S. markets. (a) DMA segmentation based on income and population density, showing systematic differences in sales efficiency. (b) Geographic distribution of alcohol expenditure per capita, highlighting regional clustering in consumption intensity.

Overall, the exploratory analysis demonstrates that DMA-level socioeconomic structure—particularly income, population density, and expenditure patterns—plays a significant role in shaping sales efficiency. These findings motivate the inclusion of contextual covariates in the hierarchical Bayesian model to ensure robust and unbiased estimation of media effectiveness across markets.

5 Model Architecture

This section presents the end-to-end architecture of the proposed hierarchical Bayesian Media Mix Modeling (MMM) framework, illustrating how raw inputs are transformed into probabilistic revenue predictions and interpretable media contributions. The architecture is designed to reflect realistic advertising dynamics while enabling principled uncertainty quantification and information sharing across markets.

Figure 5 provides a high-level overview of the modeling pipeline. The system is composed of four primary components: input ingestion, nonlinear media transformations, hierarchical parameter modeling, and probabilistic outcome generation.

Media spend inputs (X_m) first undergo nonlinear transformations to capture core advertising response mechanisms. Adstock transformations parameterized by decay rates (α) model delayed and persistent effects of media exposure, while Hill-type saturation functions characterized by slope and half-saturation parameters represent diminishing marginal returns at higher spend levels. These transformations ensure that the effective media signal reflects observed real-world response behavior rather than raw expenditure.

Contextual and control variables (Z), including demographic, economic, and seasonal factors, enter the model alongside transformed media inputs. These variables influence both baseline demand and media effectiveness through hierarchical parameterization. Media effect coefficients (β) and control effects (τ) are modeled hierarchically, with population-level priors (β_μ, β_σ) enabling partial pooling across markets. This structure allows data-sparse DMAs to borrow strength from structurally similar markets while preserving meaningful heterogeneity.

The transformed media signals and contextual effects are combined within a probabilistic likelihood model. Revenue outcomes are assumed to follow a Gaussian likelihood with noise scale parameter (σ), enabling explicit modeling of residual uncertainty. Posterior inference yields full predictive distributions rather than point estimates.

The final outputs include predicted revenue, channel-level contributions, and elasticity estimates, all accompanied by uncertainty intervals. This architecture provides a transparent and extensible framework for scenario analysis, budget optimization, and region-specific media planning under uncertainty.

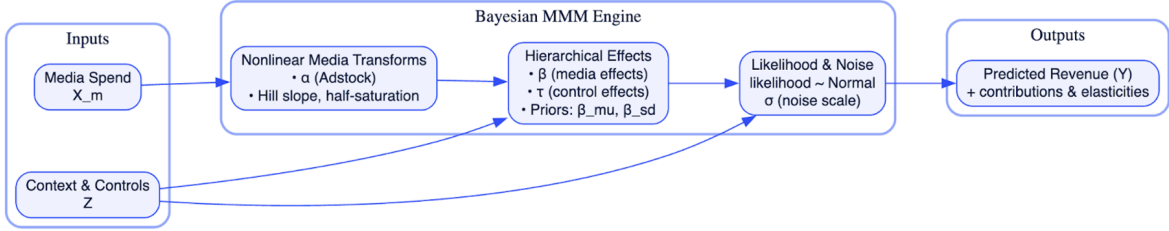


Figure 5: Hierarchical MMM Architecture

6 Model Development

6.1 Modeling Strategy and Rationale

Model development followed a structured progression to balance realism, interpretability, and statistical stability. Initial pooled linear and regularized regressions were used to assess baseline relationships between media spend and sales. While these models confirmed positive associations for major channels, their reliance on linear assumptions and global parameterization produced unstable estimates across markets, particularly for smaller or low-spend DMAs.

To address these limitations, we developed a Bayesian Media Mix Model with nonlinear response functions and extended it to a hierarchical Bayesian framework. This formulation allows key parameters to vary across DMAs while enabling partial pooling across similar markets, improving estimation stability in the presence of pronounced heterogeneity in income, population density, demographics, and market size.

6.2 Feature Engineering and Media Transformations

All sales and media variables were converted to per-capita terms, and the outcome variable was log-transformed to stabilize variance. Temporal controls capturing trend and seasonality were included to model baseline demand independent of media exposure.

Media inputs were transformed using adstock to capture delayed effects and Hill-style saturation functions to model diminishing returns. As shown in Figure 6, Linear Television and Digital Video exhibit gradual saturation with sustained carryover, Social media saturates quickly at lower spend levels, and Display shows a flatter response. These response patterns informed prior specification and hierarchical pooling decisions in the final model.

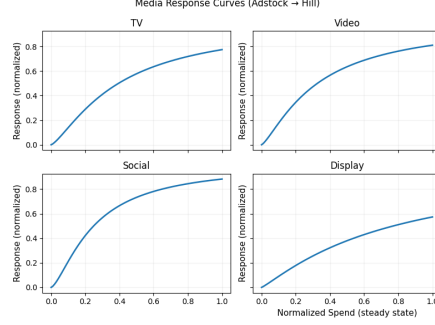


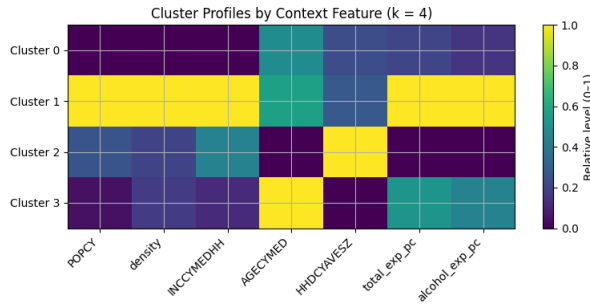
Figure 6: Estimated nonlinear media response functions after adstock and saturation transformations, illustrating channel-specific carryover and diminishing returns behavior.

6.3 Contextual Clustering and Hierarchical Structure

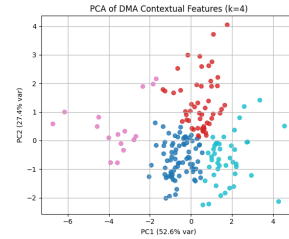
To capture structural variation across markets, DMAs were clustered using standardized socioeconomic and behavioral features, including population, density, median income, age composition, household size, total expenditure per capita, and alcohol expenditure per capita. Principal Component Analysis (PCA) was used to visualize market differentiation, with the first two components explaining a substantial share of total variance.

The PCA projection reveals clear separation into four contextual clusters ($k = 4$), indicating that DMAs form distinct market segments rather than a homogeneous population. Cluster profiles show systematic differences in income, density, age structure, and spending behavior: high-income, high-density clusters exhibit elevated total and alcohol expenditures per capita, while lower-income, rural clusters display reduced consumption and distinct demographics.

These clusters inform the hierarchical modeling structure by guiding partial pooling decisions. Rather than enforcing uniform parameter distributions across all DMAs, the hierarchical framework allows markets within similar contextual clusters to share information, improving estimation stability while preserving meaningful regional differences.



(a) Cluster Profiles by Context Features



(b) PCA Projection of DMA Clusters

Figure 7: DMA market segmentation using contextual features ($k = 4$). (a) Normalized cluster profiles highlighting differences in population, density, income, age structure, and expenditure patterns. (b) PCA projection illustrating the separation and overlap of DMA clusters in reduced-dimensional space.

6.4 Cluster-Level Media Mix Model Specification

Subsequent to the identification of four contextual clusters, we executed a cluster-level hierarchical Media Mix Model to achieve a balance between geographic flexibility and statistical stability. Instead of estimating completely independent parameters for each of the 188 DMAs, this specification aggregates DMAs within the same contextual cluster and estimates common media response parameters at the cluster level.

This design decision was driven by two pragmatic factors. Initially, numerous DMAs demonstrate sparse or irregular media activity, resulting in instability when estimating fully local nonlinear response parameters. Furthermore, the contextual clustering analysis indicated significant similarities in demographic composition, income levels, population density, and expenditure behavior across the identified clusters. This suggests that markets that are grouped together are likely to exhibit similar responses to media stimuli.

In the specification at the cluster level, the parameters for adstock and saturation are modeled specifically for each cluster. This approach permits variations in carryover strength and diminishing returns across different clusters, while maintaining consistency within each individual cluster. This framework significantly diminishes parameter dimensionality in comparison to a comprehensive DMA-level model and enhances the sampling characteristics for slope and half-saturation parameters. The cluster-level model functions as an intermediary level of detail, situated between national and entirely local models, effectively preserving significant heterogeneity while ensuring robustness.

6.5 DMA-Level Modeling and Hierarchical Constraints

We additionally investigated a comprehensive DMA-level hierarchical MMM to evaluate the potential for enhanced local flexibility to further optimize model performance. In this specification, media response parameters exhibit variability at the individual DMA level, employing hierarchical priors to regularize estimation and reduce the risk of overfitting.

In practical applications, DMA-level modeling presents considerable challenges. A considerable number of DMAs exhibit constrained variability in media expenditure, recurrent intervals of zero spending, or brief effective time series subsequent to the implementation of validation holds. The presence of these characteristics complicates the reliable estimation of DMA-specific nonlinear response parameters, especially for weaker channels like Display. Diagnostic evaluations indicated an increased posterior uncertainty concerning the slope and half-saturation parameters at the DMA level, despite the implementation of robust regularization techniques.

The observations contributed to a considered modeling decision. The final modeling strategy prioritizes cluster-level pooling, utilizing DMA-level deviations solely when substantiated by the data, rather than depending solely on fully DMA-specific parameters. This methodology maintains local variation while mitigating instability and overfitting in markets with sparse observations, thereby ensuring that parameter estimates are interpretable and appropriate for forecasting and budget optimization.

6.6 Training, Validation, and Bayesian Diagnostics

A rolling holdout validation strategy was employed to preserve temporal ordering, with the most recent weeks for each DMA reserved as a validation set to emulate realistic forecasting conditions and prevent information leakage. Parameter tuning prioritized stability and sampling behavior rather than exhaustive grid search, with prior distributions refined iteratively based on posterior

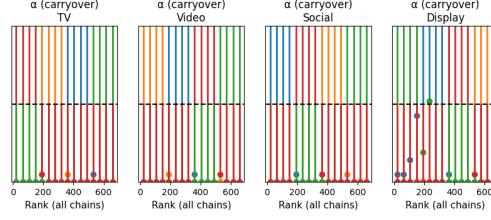


Figure 8: Carryover (α)

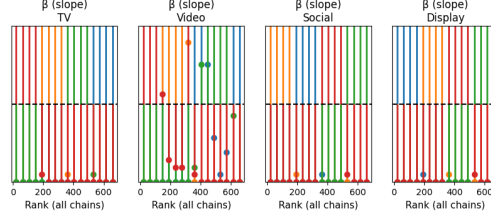


Figure 9: Slope (β)

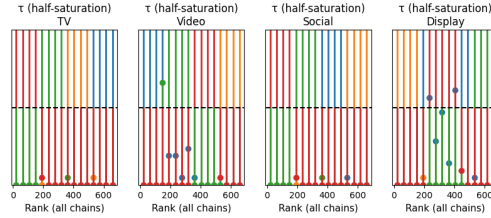


Figure 10: Half-saturation (τ)

diagnostics. Rank plots for the carryover (α), slope (β), and half-saturation (τ) parameters showed well-mixed chains with no evidence of systematic rank separation. Greater variability in slope and saturation estimates—particularly for Display—reflects inherent parameter uncertainty rather than sampling deficiencies, indicating satisfactory convergence.

7 Performance and Results

7.1 Comprehensive Predictive Precision

Model performance was evaluated using out-of-sample validation metrics. Figure 11 illustrates the comparison between predicted and observed sales values during the validation period. The hierarchical Bayesian model achieves an out-of-sample R^2 of approximately 0.77, indicating robust predictive performance given the substantial heterogeneity and noise present in the data.

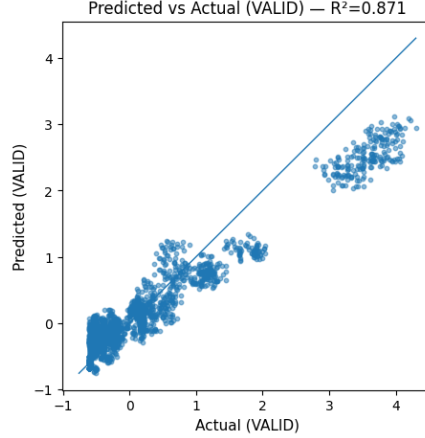


Figure 11: validation fit

Predicted values exhibit a strong overall correspondence with observed sales across most of the distribution. However, dispersion increases at higher sales levels, reflecting greater uncertainty in large, high-volume Designated Market Areas (DMAs) relative to smaller markets.

7.2 Validation Behavior at the DMA Level

To gain deeper insight into localized model behavior, validation-period time series were examined for selected DMAs. These plots show that predicted values closely track the underlying level and trend of observed sales, while appearing smoother than the actual series. This smoothing effect is particularly evident during periods of sharp week-to-week fluctuations.

This behavior is an intentional consequence of the hierarchical model structure. Partial pooling improves estimate reliability by borrowing strength from structurally similar markets, thereby reducing sensitivity to transient shocks or isolated anomalies. While this may lead to mild underprediction during extreme spikes, it substantially improves generalization and robustness across the full set of DMAs.

7.3 Error Patterns and Interpretation

The model exhibits a tendency to underestimate extreme values, especially in high-income, high-density markets during periods of elevated demand. This behavior reflects the inherent bias–variance tradeoff in hierarchical modeling. Rather than closely fitting rare extreme observations, the model shrinks estimates toward the population mean, prioritizing stability over precise replication of short-term volatility.

In contrast, lower- and mid-volume DMAs display closer alignment between predicted and observed values, suggesting that the hierarchical framework is particularly effective at stabilizing estimates in markets characterized by sparse or irregular data.

7.4 Comparison Across Modeling Approaches

Relative to national-level baseline models, the hierarchical specification demonstrates more consistent performance across the full distribution of markets. While national models align well with aggregate trends, they often fail to generalize effectively at the local level. The hierarchical model

accepts modest tradeoffs in peak-level fit in exchange for improved reliability, interpretability, and applicability across diverse market contexts.

8 Conclusion and Next Steps

8.1 Key Takeaways

This project illustrates that geographic markets exhibit systematic variations in both baseline demand and their responsiveness to advertising efforts. The factors of income, density, age composition, and expenditure patterns significantly influence the efficiency of sales and the effectiveness of media strategies. Considering all DMAs as uniform may conceal these distinctions and result in unreliable or erroneous recommendations.

The integration of nonlinear media response functions within a hierarchical Bayesian framework, alongside contextual clustering, results in a model that offers a more realistic and robust methodology for media measurement. The model effectively accounts for carryover and saturation effects, enhances the stability of estimates in sparse markets, and demonstrates superior out-of-sample performance compared to more straightforward alternatives.

8.2 Limitations

The analysis is based on observational data and may be influenced by potential confounding factors and measurement inaccuracies. Although hierarchical pooling enhances stability, it may diminish the system’s responsiveness to sudden local disturbances. Certain contextual variables might serve as inadequate proxies for consumer behavior, and the lack of specific controls, including pricing or promotion data, could constrain the accuracy of full attribution.

8.3 Future Improvements

Future research may investigate alternative clustering methodologies, integrate explicit spatial correlation frameworks, or permit media response parameters that are specific to individual clusters. Enhancing the model to incorporate automated retraining and scenario simulation would significantly augment its practical utility. Where applicable, the incorporation of experimental or quasi-experimental signals may enhance causal interpretation and bolster confidence in decision-making.

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